Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Practical experimentation is fundamental to physics education, yet many schools in rural Sri Lanka lack adequate laboratory infrastructure, equipment, and technical support. Students are consequently limited to theoretical learning without sufficient practical exposure, restricting the development of scientific reasoning and conceptual understanding. Most existing digital learning platforms treat experiment simulations, quiz assessments, and performance monitoring as separate features, without integrating them into a unified adaptive learning environment.

This research proposes a mobile-based virtual physics laboratory application designed for Grade 6–11 students in rural learning environments. The system enables students to perform interactive physics experiments, complete lesson-based quizzes, and engage in gamified learning activities while receiving guidance from an AI-powered educational chatbot. A key contribution of the proposed system is the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module. This module integrates quiz results, virtual experiment performance, and lesson progression to analyze student learning, estimate student levels, detect weaknesses, and generate personalized feedback within the mobile virtual physics laboratory.

Each lesson is divided into sub-lessons assessed through both quizzes and virtual experiments. Quizzes are intelligently generated using MCQs and structured questions aligned with the Grade 6–11 curriculum, while experiments evaluate practical skills. The module displays separate scores for each sub-lesson and identifies weak areas. When low performance is detected, the system immediately assigns a personalized learning path containing explanations, guided examples, experiment procedures, and video materials to support deeper understanding of the weak sub-lesson. After engaging with the learning path, the student reattempts the quiz and experiments for that sub-lesson. If the student still performs poorly after reattempting, the system assigns additional external resources such as YouTube videos and reference materials for further knowledge building. Shuffled quiz retakes and structured progression prevent students from advancing without completing the current sub-lesson.

The system will be implemented using Flutter for mobile application development, Firebase Authentication for secure user management, Cloud Firestore for storing learning data, and a Python-based backend for analytics processing and adaptive feedback generation. The expected outcome is an accessible and intelligent mobile learning platform that enhances physics conceptual understanding while providing weakness detection, personalized feedback, and targeted remediation support for students in resource-limited educational environments.

Keywords: Virtual Physics Laboratory, Mobile Learning, Intelligent Learning Analytics, Adaptive Feedback, Weak Student Remediation, Personalized Instruction, Physics Education

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LIST OF ABBREVIATIONS

Abbreviation	Description
MCQ	Multiple Choice Question
API	Application Programming Interface
FR	Functional Requirement
NFR	Non-Functional Requirement
UI	User Interface
NoSQL	Non-Relational Structured Query Language
SDK	Software Development Kit
JSON	JavaScript Object Notation
LMS	Learning Management System
STEM	Science, Technology, Engineering, and Mathematics

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1. INTRODUCTION

1.1 Background and Context

Physics education requires both theoretical understanding and practical experimentation to help students develop scientific reasoning and conceptual knowledge. However, many schools in developing regions lack adequate laboratory infrastructure, equipment, and trained instructors, which limits students' opportunities to perform experiments and observe scientific phenomena directly [1]. This challenge is particularly evident in rural educational environments where resources and laboratory facilities are limited.

To address these limitations, digital learning technologies such as virtual laboratories have emerged as promising solutions for improving access to practical science education. Virtual laboratory systems allow students to perform simulated experiments and interact with scientific concepts through digital environments without requiring physical laboratory equipment [2]. Research has shown that virtual experiments can enhance students' conceptual understanding and provide flexible learning opportunities that overcome time and location barriers in traditional education systems [1].

In addition to experiment simulations, modern educational platforms increasingly incorporate mobile learning technologies, quizzes, gamified activities, and intelligent tutoring systems to enhance student engagement and support interactive learning experiences. Mobile learning technologies allow students to access educational resources anytime and anywhere using smartphones or tablets, making them particularly useful for learners in remote or underserved areas [3]. Gamification techniques such as challenges, rewards, and progress indicators have also been shown to increase student motivation and participation in STEM education [5]. Furthermore, artificial intelligence technologies such as educational chatbots can provide automated explanations and learning guidance to support students during their learning process [10].

Despite these advancements, many existing educational systems treat experiment interaction, quiz evaluation, and learning analytics as separate components. Consequently, student performance data becomes fragmented and does not provide a comprehensive view of a student's conceptual understanding across lessons and activities [1]. To overcome this limitation, the proposed research aims to develop a mobile-based virtual physics laboratory platform that integrates experiments, quizzes, gamified learning activities, and AI-assisted learning support within a unified learning environment. Within this system, the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module plays a central role in monitoring student learning progress and generating personalized feedback based on integrated learning analytics.

1.2 Literature Review

Recent research highlights the increasing role of virtual laboratories and simulation technologies in science education. Virtual laboratory environments allow students to explore scientific phenomena through interactive simulations and digital experiments that support conceptual visualization and practical understanding of complex scientific concepts [3]. Studies have demonstrated that integrating virtual experiments into science education can significantly improve students' engagement and conceptual learning outcomes [11]. In particular, virtual experiments allow students to repeat experiments multiple times, explore scientific relationships interactively, and learn at their own pace in a safe digital environment [2].

Another important research area related to this study is learning analytics, which focuses on collecting and analyzing educational data to better understand student learning behavior and improve learning outcomes. Learning analytics enables educators and learning systems to monitor student engagement, identify learning patterns, and detect potential learning difficulties based on interaction data generated during educational activities [1]. Educational data mining techniques further support this process by analyzing large datasets generated from digital learning platforms to discover meaningful patterns in student learning behavior and academic performance [7]. Researchers have emphasized that learning analytics systems can help monitor student progress and support data-driven decision making in educational environments [6]. Research has also highlighted that personalised learning supported by big data and artificial intelligence presents significant opportunities for improving educational outcomes [9]. Recent work has also demonstrated that integrating AI-based performance prediction with learning analytics significantly improves student outcomes in online courses [15], and that equity-focused learning analytics in physics education requires structured guidance to be effective [17].

In addition to analytics-based evaluation, recent research has explored the role of artificial intelligence in providing personalized learning support. AI-based educational systems can analyze student responses and provide automated feedback to guide learners toward better conceptual understanding [10]. Studies have shown that leveraging AI in e-learning through personalized learning and adaptive assessment enhances student engagement and academic performance [16]. Recent studies have also demonstrated that combining interactive learning environments with AI-generated feedback mechanisms can significantly improve student learning outcomes and engagement in physics laboratory environments [14].

Mobile learning technologies also play a significant role in expanding access to education. Mobile learning platforms enable students to access educational content through smartphones and tablets, providing flexible learning opportunities that are particularly beneficial for students who cannot access traditional classroom environments [4]. However, many existing mobile learning applications focus mainly on content delivery or interactive activities without integrating structured performance evaluation mechanisms that combine theoretical assessments and practical experiment outcomes within a unified learning model.

1.3 Research Gap

Despite the increasing use of virtual laboratories and digital learning platforms in science education, several limitations remain in current educational systems. Many existing virtual laboratory platforms primarily focus on providing experiment simulations without evaluating how effectively students understand the underlying scientific concepts through those experiments. Although virtual experiments allow students to visualize scientific phenomena and interact with simulated environments, they often lack mechanisms to systematically monitor student learning performance across lessons and activities [3].

Another limitation in existing educational systems is that theoretical assessments and practical experiment activities are often implemented as independent learning features. Many digital learning platforms evaluate theoretical knowledge through quizzes, while experiment-based systems focus mainly on simulation interaction without linking experiment outcomes to student assessment data. As a result, these systems provide only partial insight into students' conceptual understanding and learning progress [8].

Furthermore, many existing learning analytics platforms are designed for large-scale learning management systems used in higher education environments. These systems typically require advanced technological infrastructure and may not be suitable for school-level learning environments or rural educational contexts where technological resources are limited [13]. Although virtual experiments and mobile learning platforms such as PhET simulations improve accessibility and engagement in science education [12], relatively few systems integrate experiment activities, quiz evaluation, learning analytics, and personalized feedback within a single mobile learning environment.

Recent research also highlights the importance of providing individualized feedback during experimental learning activities to improve conceptual understanding and learning performance in physics education [14]. Studies further demonstrate that integrating AI-based performance prediction with learning analytics improves student outcomes [15], and that personalized learning through adaptive assessment is essential for effective e-learning [16]. However, most existing virtual laboratory systems do not incorporate automated personalized feedback mechanisms based on integrated analysis of quiz results, experiment interactions, and lesson-based learning progress.

Therefore, a clear research gap exists in the development of an integrated mobile learning system that combines user management, quiz evaluation, virtual experiment performance analysis, and learning analytics to generate personalized learning feedback within a virtual physics laboratory environment. As noted by Grimm et al. [17], learning analytics in physics education still lacks structured guidance for equity-focused decision-making, reinforcing the need for adaptive systems. The proposed Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module addresses this gap by integrating quiz results, virtual experiment performance, and lesson progression to analyze student learning, estimate student levels, detect weaknesses, and generate personalized feedback in a mobile virtual physics laboratory.

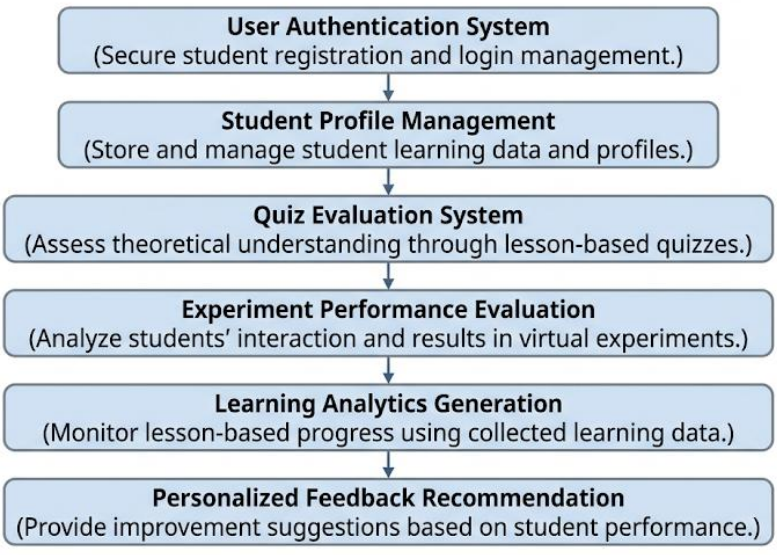
2. OBJECTIVE

2.1 Main Objective

To design and implement an **Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module** that detects weak students through quiz and experiment performance, assigns personalized learning paths for deeper learning, and provides additional resources for students who remain weak after reattempting assessments, within a mobile virtual physics laboratory application.

2.2 Specific Objectives

- Monitor student engagement in quizzes and virtual experiments at sub-lesson and topic levels.
- Generate intelligently designed quizzes covering Grade 6–11 curriculum content using MCQs and structured questions.
- Analyze student performance to detect weak areas in both theoretical and practical understanding.
- Generate personalized feedback messages and display improvement suggestions on the student home page based on detected weak areas.
- Assign personalized learning paths to weak students based on low quiz and experiment performance in specific sub-lesson areas, and enable reattempting of assessments after completing the learning path.
- Provide additional external resources such as YouTube videos and reference materials to students who continue to show low performance after completing the learning path and reattempting the quiz and experiment.
- Support repeated practice through shuffled quiz variations to encourage deeper conceptual understanding.
- Enforce structured learning progression by ensuring students complete each sub-lesson before advancing to the next..



3. METHODOLOGY

3.1 System Design Approach

The proposed system is designed as a **mobile learning platform** where students interact with physics learning materials through experiments, quizzes, and gamified learning activities. The Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module acts as the core analytical component of the system.

The system performs four main processes:

1. **Student Performance Monitoring**
2. **Weakness Detection and Analysis**
3. **Adaptive Feedback Generation**
4. **Personalized Remediation and Instructional Support**

These processes work together to monitor student learning behavior and generate personalized learning recommendations.

3.2 Sub-Lesson Based Performance Monitoring

The system is developed as part of the mobile virtual physics laboratory to manage student learning progress and evaluate performance in a more adaptive and supportive way. Each main physics lesson is divided into sub-lessons. For example, a lesson such as Force may contain sub-lessons like 1.1, 1.2, and 1.3. Each sub-lesson includes two assessment methods: quizzes and virtual experiments. The quizzes are intelligently generated for each lesson and sub-lesson, covering Grade 6–11 curriculum content, and include a combination of Multiple Choice Questions (MCQs) and structured questions to assess students' theoretical understanding at different cognitive levels.

For each sub-lesson, the app displays separate marks for quiz performance and experiment performance, such as Quiz = 75% and Experiment = 30%, and then uses these results to identify student strengths and weak areas. The quizzes measure students' theoretical knowledge, while the experiments evaluate practical understanding. Based on these two assessment results, the system analyzes performance at both sub-lesson level and main topic level.

- Correct and incorrect quiz questions per sub-lesson
- Virtual experiment performance and step completion status
- Identified weak theory and practical areas
- Personalized feedback messages and home page suggestions
- Experiment performance records
- Adaptive remediation resources and structured learning progression

All performance data is stored securely in Cloud Firestore and displayed through the student dashboard as feedback messages and learning suggestions on the home page.

3.3 Detailed Performance Analysis and Weakness Detection

A key feature of this module is its ability to provide detailed performance analysis rather than only showing marks. The system identifies which quiz questions were answered correctly and which had low scores, helping students understand their strong and weak theory areas. In the experiment section, the app evaluates practical performance and identifies which parts the student performed successfully and which parts need improvement. This analysis is summarized and presented through feedback messages and a suggestion section on the home page.

Theory Assessment – Quiz Analysis

Students complete intelligently generated quizzes for each sub-lesson, covering Grade 6–11 curriculum content. Each quiz includes MCQs and structured questions to assess theoretical understanding at different cognitive levels. The system evaluates answers, identifies incorrectly answered questions, and stores results with incorrect answer details and topic tags for further analysis.

Practical Assessment – Experiment Evaluation

Students perform virtual experiments within the mobile application. The system evaluates their practical performance and identifies which parts of the experiment were completed successfully and which parts need improvement. This analysis helps detect weak practical understanding at the sub-lesson level.

Lesson Progression and Structured Learning Enforcement

The module enforces structured learning progression by ensuring that students cannot move to the next sub-lesson without completing the current one. Lesson completion status is tracked and used along with quiz and experiment results to build a complete picture of the student's learning progress.

All collected data is stored securely in **Cloud Firestore**, allowing real-time synchronization between the mobile application and backend services.

3.4 Adaptive Feedback Recommendation for Weak Students

The module supports **adaptive feedback recommendation** for weak students. If a student receives a very low score in the experiment part of a sub-lesson, the system recognizes weak practical understanding and provides additional support such as simple explanations, guided examples, correct experiment procedures, and video-based learning materials.

If the student receives a low score in the quiz part, the system identifies incorrectly answered questions and provides **step-by-step explanations**, concept clarification, and supporting learning resources to improve theoretical understanding.

The module also supports repeated practice. When a student attempts a quiz again, the system presents a different shuffled set of MCQ and structured questions, ensuring variation and reducing memorization-based answering. This encourages deeper conceptual understanding of the lesson content.

Once the student engages with the assigned learning path and completes the provided resources, the system encourages the student to reattempt the quiz and experiment for that specific weak sub-lesson to demonstrate improved understanding. If the student still receives a low score after completing the learning path and reattempting the assessment, the system recognizes persistent weakness and assigns

additional external learning resources such as YouTube videos and reference materials specifically selected for that weak sub-lesson to provide further support for deeper knowledge building.

This approach allows the system to evaluate both theoretical knowledge and practical experiment performance together, generating targeted adaptive support for each student rather than treating assessments as separate learning activities.

3.5 Personalized Instruction and Feedback Generation

Based on quiz and experiment performance analysis, the system generates personalized instructional feedback and assigns structured learning support to students based on their identified weak areas. This transforms the mobile virtual physics laboratory into an intelligent learning support system that not only evaluates performance but also detects weaknesses and delivers targeted instructional support to enhance physics learning outcomes.

The personalized feedback and instructional support follow a rule-based decision model that evaluates quiz and experiment scores at sub-lesson level and responds in two distinct ways: immediate feedback messages and a personalized learning path assignment for weak students.

Immediate Feedback

After each quiz or experiment attempt, the system generates a feedback message specific to the student's performance in that sub-lesson. These feedback messages include:

- Sub-lesson quiz and experiment performance marks (e.g., Quiz = 45%, Experiment = 30%)
- Identification of which quiz questions were answered incorrectly, and which experiment steps were performed poorly
- A brief explanation of the weak concepts or practical areas detected
- A suggestion displayed on the home page encouraging the student to review the weak areas

The student cannot progress to the next sub-lesson until the current sub-lesson is satisfactorily completed, ensuring structured learning progression.

Personalized Learning Path Assignment for Weak Students

When a student receives a low score in the quiz or experiment of a sub-lesson, the system immediately recognizes the weak area and assigns a personalized learning path specifically designed for that student based on the weak lesson or sub-lesson identified. The system notifies the student and prompts them to accept the recommended learning path.

For example, if a student scores low in the Force lesson — Sub-lesson 1.1, the system detects this weakness and assigns a learning path specifically for Force 1.1. The student receives a notification suggesting the learning path and must accept it to proceed with the additional support.

Once the student accepts the learning path, it provides:

- Simple explanations of the weak theory concepts with worked examples relevant to the specific weak sub-lesson
- Step-by-step concept clarification for incorrectly answered quiz questions in that sub-lesson

- Correct experiment procedures with guided examples for poorly performed experiment steps
- Video-based learning materials aligned to the specific weak sub-lesson content
- Additional external learning resources for deeper understanding of the weak lesson topic

This learning path is personalized to each individual student based on their exact weak areas detected from both quiz and experiment performance. It is assigned immediately when weakness is detected, not after repeated attempts, ensuring that every student who struggles receives targeted instructional support at the right time. Once the student completes the learning path and engages with all provided resources, the system encourages the student to reattempt the quiz and experiment for that specific weak sub-lesson to demonstrate improved understanding. If the student still performs poorly after completing the learning path and reattempting the assessment, the system recognizes persistent weakness and assigns additional external resources such as YouTube videos and reference materials specifically selected for that weak sub-lesson, providing escalating instructional support until the student's understanding of the specific sub-lesson improves.

3.6 Learning Analytics Visualization

To help students understand their learning progress, the system displays performance results and feedback through the student dashboard in the mobile application. Flutter chart libraries and progress indicators are used to display information such as:

- Sub-lesson quiz and experiment marks (e.g. Quiz = 75%, Experiment = 30%)
- Identified strong and weak theory and practical areas per sub-lesson
- Personalized feedback messages and home page suggestions
- Lesson progression status and sub-lesson completion indicators

These visual indicators provide students with a clear overview of their quiz and experiment performance, weak areas, and learning progress across all physics sub-lessons and topics.

3.7 System Implementation Technologies

Several technologies are used to implement the proposed system:

- **Flutter** – Mobile application development framework used to create the student interface and interactive learning screens.
- **Firebase Authentication** – Used for secure student registration and login management.
- **Cloud Firestore** – Cloud-based NoSQL database used to store student profiles, quiz results, experiment records, and learning progress data.
- **Python Backend (FastAPI / Flask)** – Used to process learning analytics, evaluate performance data, and generate personalized feedback.
- **Flutter Chart Libraries** – Used to visualize learning analytics data through interactive graphs and progress indicators.

3.8 System Evaluation

The proposed system will be evaluated using both **functional testing and usability evaluation**.

Functional testing ensures that the system correctly performs tasks such as authentication, quiz evaluation, experiment performance tracking, and feedback generation.

Usability testing will assess how effectively students can interact with the application and understand the generated feedback and learning analytics dashboard.

Performance evaluation metrics include:

- accuracy of quiz score calculation
- correctness of experiment performance evaluation
- response time of analytics processing
- usability of the learning dashboard
- effectiveness of personalized feedback recommendations

These evaluation metrics will help determine whether the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module successfully supports structured monitoring of student learning progress within the mobile virtual physics laboratory environment.

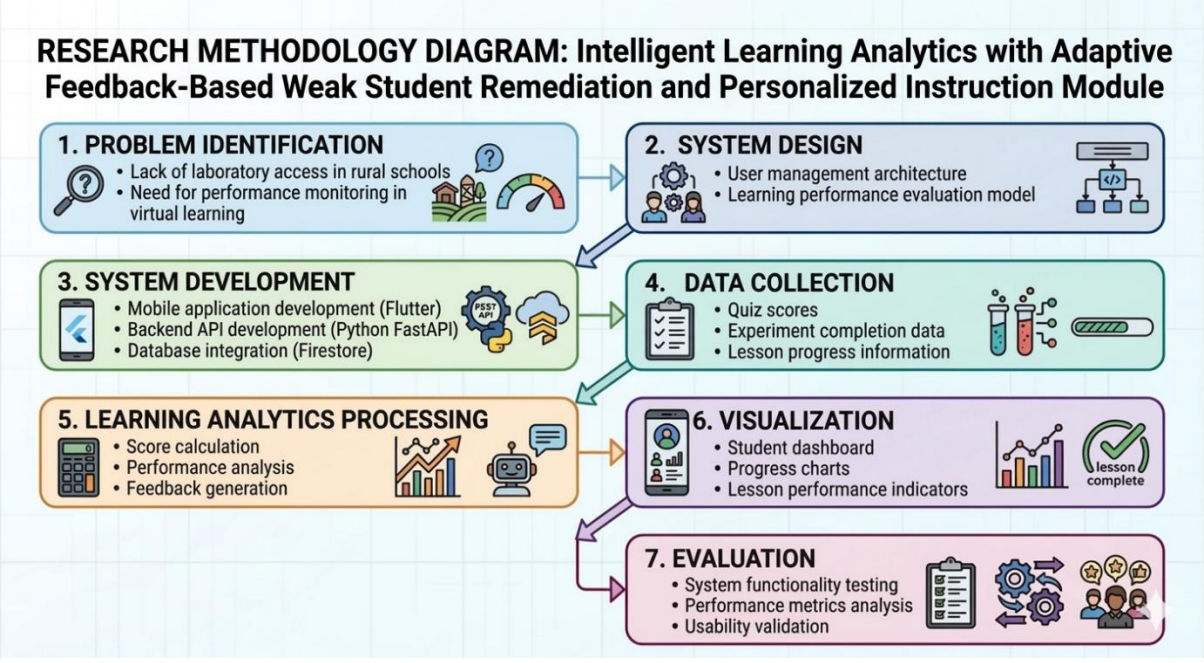


Figure 1-Methodology Diagram

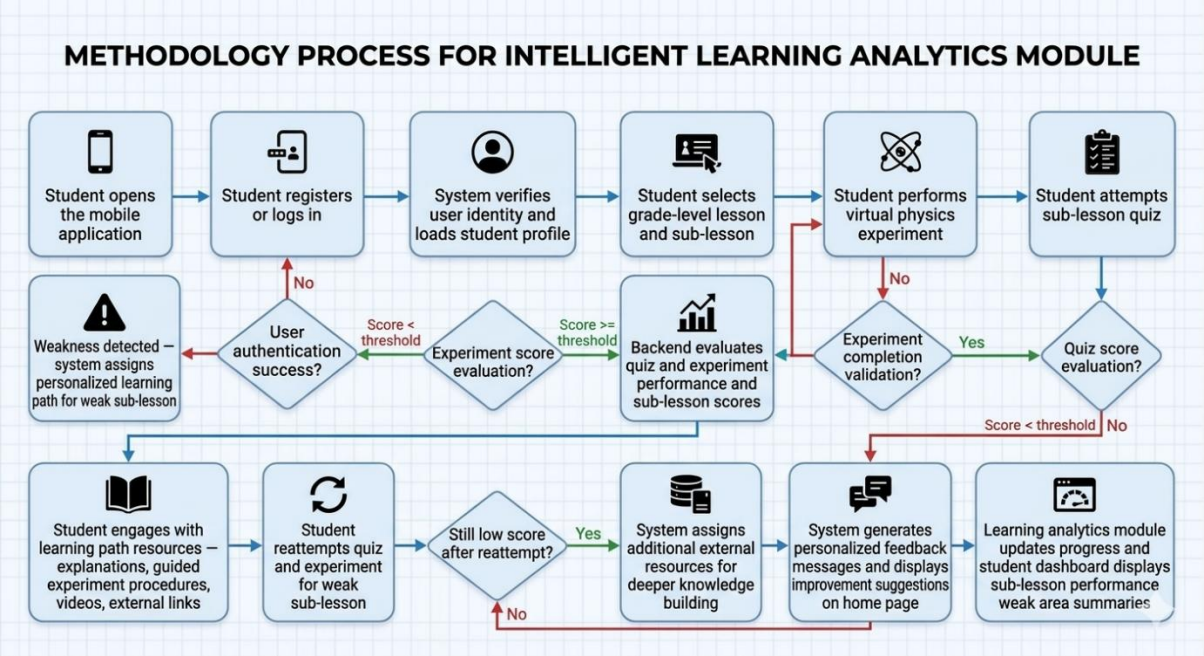


Figure 2-Methodology Process

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed solution is a mobile-based virtual physics laboratory application designed for Grade 6–11 students in rural Sri Lanka. The system combines four main functional components to support interactive and structured physics learning. These are the Grade-Adaptive AI Chatbot with Structured Learning Navigation, the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module, the Interactive Physics Experimentation and Learning Module, and the Intelligent Content Retrieval and Gamified Learning Generation Module. Together, these components create a unified mobile learning environment where students can access grade-based physics lessons, perform practical activities, answer quizzes, receive AI assistance, and monitor their lesson-wise learning progress.

At a high level, the system is organized into three layers. The first is the presentation layer, which is the Flutter mobile application interface used by students. This layer includes screens for login and registration, lesson selection, quizzes, virtual experiments, educational games, chatbot interaction, the learning progress dashboard, the personalized learning path interface for weak students, and the reattempt screen for students who have completed their assigned learning path. The second is the application logic layer, which contains the backend services responsible for quiz evaluation, experiment performance analysis, weakness detection, adaptive feedback generation, personalized learning path assignment, learning path completion tracking, reattempt evaluation, additional external resource assignment for persistently weak students, content retrieval, chatbot processing, and learning analytics generation. The third is the data layer, which stores student profiles, lesson data, quiz scores, experiment results, detected weak areas, assigned learning paths, learning path completion status, reattempt results, additional resource assignments, game records, and progress information using Cloud Firestore and related backend data services.

The Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module, which is the individual component of this research, plays a central

role in the system. It contains five main parts: user authentication, profile management, performance tracking, learning analytics generation, and personalized learning path assignment for weak students. The user authentication part manages registration, login, Google sign-in, and guest access through Firebase Authentication. The profile management part stores learner details such as name, grade level, and registration information. The performance tracking part records quiz results and virtual experiment outcomes for each sub-lesson, identifying which quiz questions were answered incorrectly and which experiment steps were performed poorly. The learning analytics part processes these records and generates sub-lesson level performance indicators, quiz and experiment scores, weakness summaries, and personalized feedback messages displayed on the student dashboard and home page suggestion section. The personalized learning path assignment part detects weak students based on low quiz or experiment scores in a specific sub-lesson and immediately assigns a targeted learning path containing simple explanations, guided examples, correct experiment procedures, video-based learning materials, and external resources to support deeper understanding of the weak sub-lesson content. Once the student completes the assigned learning path and engages with the provided resources, the system encourages the student to reattempt the quiz and experiment for that specific sub-lesson to demonstrate improved understanding. If the student still receives low marks after completing the learning path and reattempting the assessment, the system recognizes persistent weakness and assigns additional external learning resources specifically selected for that weak lesson topic to provide further support for deeper knowledge building. This escalating support mechanism ensures that every weak student receives continuous instructional support until their understanding of the specific sub-lesson improves.

The system components interact closely with one another. When a student logs into the application, the User Authentication part validates the identity and loads the student profile. The Learning Navigation and AI Chatbot component uses the student's grade level to guide the learner to appropriate lessons and provide level-appropriate explanations. The Interactive Physics Experimentation Module allows the student to perform virtual practicals and sends experiment completion and performance data to the backend. The Intelligent Content Retrieval and Gamified Learning Module provides supporting lesson content and lesson-based educational games. The Intelligent Learning Analytics Module receives quiz results and practical experiment data from these components, evaluates performance at sub-lesson level, detects weak areas, generates feedback messages, assigns personalized learning paths to weak students, tracks learning path completion, evaluates reattempt results, assigns additional external resources for persistently weak students, stores all results in the database, and updates the student's learning dashboard.

The data flow begins when the student interacts with the Flutter mobile application. Authentication requests are sent to Firebase Authentication for login validation. After successful login, the student profile is loaded from Cloud Firestore. When the learner performs a quiz or virtual experiment, the mobile application sends the activity data to the Python backend. The backend evaluates quiz answers, analyzes experiment outcomes, generates sub-lesson performance scores, detects weak areas, produces personalized feedback, and where weakness is identified, assigns a specific learning path for that student. Once the student completes the learning path and reattempts the quiz or experiment, the backend re-evaluates the new results. If the student still performs poorly, the backend assigns additional external resources for that specific weak sub-lesson. All processed results are stored in Cloud Firestore and retrieved by the dashboard and home page to display lesson-based analytics, feedback messages, learning path recommendations, and additional resource assignments, allowing the student to monitor progress and access escalating targeted support over time.

From a scalability perspective, the architecture is modular, which means each component can be extended independently without affecting the whole system. New lessons, experiments, quizzes,

learning path resources, additional external resources, and games can be added for additional topics or grade levels. Cloud Firestore supports scalable real-time data storage, while the backend can be expanded to handle more analytics features, learning path content, and escalating resource assignment logic as the user base grows. From a security perspective, the system uses secure authentication through Firebase, controlled database access, and authenticated API communication between the mobile application and backend. Student information is limited to necessary educational data, and performance records, learning path assignments, reattempt results, and additional resource assignments are stored securely to protect user privacy. This architecture therefore supports both future expansion and safe educational use.

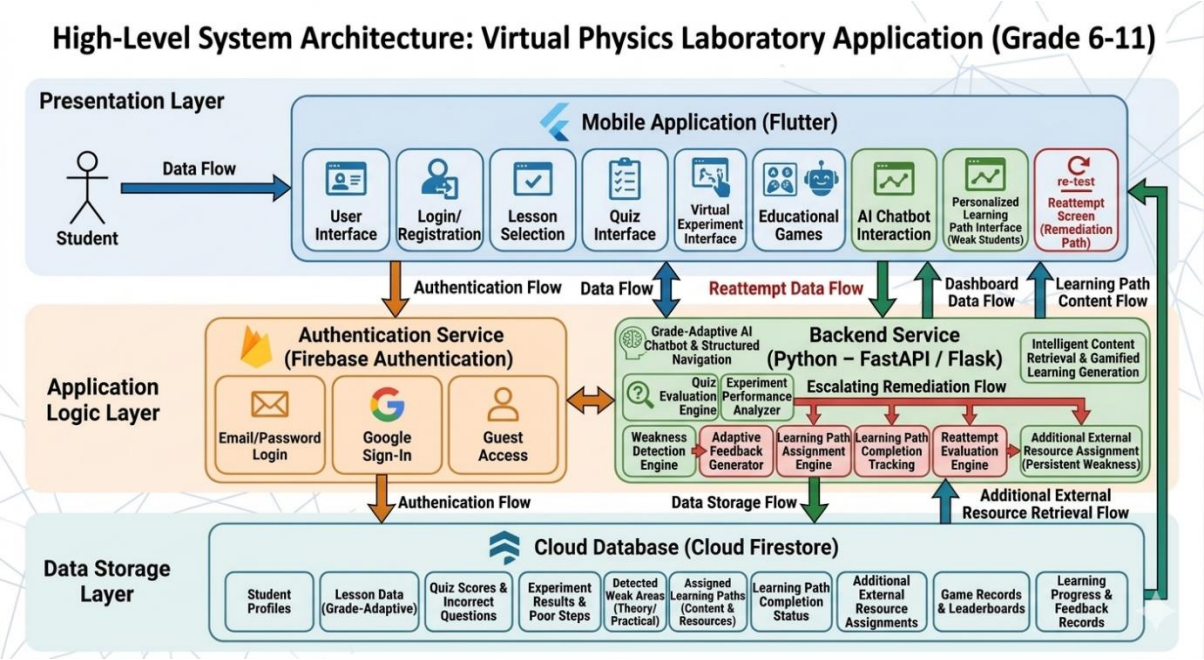


Figure 3-High-level system Archytecture

5. USER REQUIREMENTS

Stakeholder Identification

The proposed system involves several stakeholders who interact with or benefit from the mobile virtual physics laboratory application.

• **Students (Primary Users)** – Grade 6–11 students who use the mobile application to perform virtual physics experiments, answer quizzes, access learning materials, and monitor their learning progress..

• **School Administrators** – Educational institutions that may adopt the platform to improve access to laboratory-based learning for students.

• **Researchers / Developers** – Individuals responsible for developing, maintaining, and improving the system based on educational and technical requirements.

• **Educational Authorities** – Organizations responsible for curriculum development and digital education initiatives that may use the platform to support physics education in rural schools.

Functional Requirements

The system must provide the following functionalities.

1. The system shall allow students to register and create user accounts using email and passwords.
2. The system shall support secure login authentication, including Google Sign-In and guest access.
3. The system shall allow students to create and manage personal profiles, including name, grade level, and learning history.
4. The system should provide grade-based access to learning materials, including experiments, quizzes, and educational games.
5. The system shall allow students to perform virtual physics experiments through the mobile application.
6. The system shall record experiment completion status and evaluate practical performance based on task completion and accuracy of each experiment step.
7. The system shall provide intelligently generated sub-lesson quizzes covering Grade 6–11 curriculum content, including Multiple Choice Questions and structured questions to assess theoretical understanding at different cognitive levels.
8. The system shall automatically calculate and display separate quiz and experiment performance scores for each sub-lesson.
9. The system shall identify which quiz questions were answered incorrectly and which experiment steps were performed poorly for each sub-lesson.
10. The system shall detect weak theory and practical areas per sub-lesson based on quiz and experiment performance results.

11. The system shall generate personalized feedback messages and display improvement suggestions on the student home page based on detected weak areas.
12. The system shall assign a personalized learning path to weak students when low performance is detected in a specific sub-lesson, providing targeted resources to improve their knowledge in that weak area.
13. The system shall provide learning path resources including simple explanations, guided examples, correct experiment procedures, video-based learning materials, and external resources specific to the weak sub-lesson.
14. The system shall support repeated quiz attempts with shuffled question sets to encourage deeper conceptual understanding and prevent memorization-based answering.
15. The system shall allow students to reattempt the quiz and experiment for a weak sub-lesson after completing the assigned personalized learning path, to demonstrate improved understanding.
16. The system shall assign additional external resources such as YouTube videos and reference materials to students who still show low performance after completing the learning path and reattempting the quiz and experiment.
17. The system shall enforce structured learning progression by preventing students from advancing to the next sub-lesson without completing the current one.
18. The system shall store student learning data, including quiz scores, experiment outcomes, detected weak areas, assigned learning paths, and lesson progress.
19. The system shall display learning analytics and progress indicators through a student dashboard, including sub-lesson performance summaries and feedback messages.
20. The system shall allow students to track lesson-wise and sub-lesson-wise learning progress over time.
21. The system shall integrate with AI chatbot support for answering physics-related questions..

Non-Functional Requirements

The system must satisfy the following quality requirements.

Security

• The system shall implement secure authentication mechanisms for user login and registration.
• Student data shall be stored securely in the cloud database.
• Only authenticated users shall access personal learning data and dashboards.
• Communication between the mobile application and backend services shall be protected using secure APIs.

Performance

• The system shall provide fast response time for login, quiz evaluation, and data retrieval.
• The mobile application shall run efficiently on standard Android smartphones.
• Learning analytics data should be displayed without significant delays.

Usability

• The user interface shall be simple and intuitive for school students.
• Navigation between lessons, experiments, quizzes, and dashboards shall be easy to understand.
• The application shall provide clear feedback messages and learning progress indicators.

Reliability

• The system shall store learning progress data accurately.
• The system shall maintain stable operation during learning activities such as quizzes and experiments.

Scalability

• The system architecture shall support future expansion to additional lessons, experiments, and users.
• The cloud database shall allow scalable storage of student learning data.

COMMERCIALIZATION PLAN

Target Market and Customer Persona

The primary target market for the proposed system includes school students in Grades 6–11, particularly those studying physics in schools that have limited laboratory resources. The application is especially useful for students in rural and semi-urban educational environments where access to practical laboratory equipment is restricted. Secondary target users include teachers and educational institutions that wish to supplement classroom learning with digital experiments and interactive learning activities. The typical user persona is a school student who owns or has access to a smartphone and wants an interactive and accessible way to understand physics concepts and perform virtual experiments.

Value Proposition

The proposed system provides a mobile-based virtual physics learning environment that allows students to perform digital experiments, complete quizzes, and track their learning progress. Unlike traditional learning methods that rely heavily on theoretical explanations, the platform offers interactive and visual learning experiences that help students understand scientific concepts more effectively. The system also integrates learning analytics to help students monitor their progress and identify weak areas. This combination of experimentation, assessment, and progress tracking creates a more engaging and structured learning experience.

Revenue Model

The system can follow a freemium educational technology model. The basic version of the application can be provided free of charge to ensure accessibility for students, particularly in rural schools. Additional features such as advanced analytics, extended experiment libraries, premium educational content, and teacher monitoring tools could be offered through subscription-based services or institutional licenses. Educational institutions or private learning centers may subscribe to these premium features to enhance their teaching resources.

Cost Estimation

The development cost of the proposed system is relatively low because it uses open-source development frameworks and cloud services with free usage tiers. Technologies such as Flutter, Firebase Authentication, Cloud Firestore, and Python backend frameworks reduce the need for expensive software licenses. The primary costs would involve developer time, server resources for scaling the application, and maintenance activities if the system is deployed at a larger scale.

Pricing Strategy

A tiered pricing strategy can be adopted for the system. The core educational features, including basic experiments and quizzes, can remain free to encourage widespread adoption among students. Premium features such as additional experiments, advanced analytics dashboards, teacher monitoring tools, and expanded educational content can be offered through monthly or yearly subscription plans. Educational institutions may also receive discounted institutional packages that allow multiple students to access premium features.

Competitive Advantage

The proposed platform offers several advantages compared to existing digital learning systems. It combines virtual experiments, quizzes, learning analytics, and AI-assisted guidance within a single mobile application. Many existing educational tools provide only individual features such as simulations or quizzes. By integrating multiple learning components into a unified platform, the system provides a more comprehensive learning environment. Additionally, the mobile-based design makes the platform accessible to students who may not have access to traditional laboratory resources.

Intellectual Property Considerations

The system may involve intellectual property related to software architecture, experiment simulation designs, and learning analytics algorithms. These components can be protected through software copyrights and appropriate licensing agreements. If the system is expanded into a commercial educational product, the developers may consider registering trademarks for the platform name and protecting unique learning features to maintain ownership and prevent unauthorized duplication of the software.

7. BUDGET AND JUSTIFICATION

Hardware Costs

The proposed system does not require specialized hardware equipment because the application is designed to run on standard mobile devices and personal computers used for development. The development process can be carried out using existing computers available to the research team. Students who use the system will access the application through Android smartphones, which are widely available among school students. Therefore, no additional hardware purchases are required for the research prototype.

Software Licenses

The system will be developed using open-source software frameworks and free development tools, which eliminates the need for purchasing software licenses. Technologies such as Flutter for mobile development, Python frameworks (FastAPI or Flask) for backend services, and development environments like Visual Studio Code or Android Studio are freely available. Since these tools are open-source or provide free usage for development purposes, the project does not incur licensing expenses.

Cloud Services

The proposed system uses Firebase Authentication and Cloud Firestore for user authentication and data storage. These services provide free-tier usage plans that are sufficient for a research prototype and small-scale testing. The free tier allows secure user login management, storage of student profiles, quiz scores, and experiment performance data. If the system is deployed for large-scale use in the future, additional cloud resources may be required, but for the research stage, the available free cloud services are adequate.

Human Effort Cost

The system development is carried out as part of an undergraduate research project, and therefore the development work is performed by student researchers as part of their academic requirements. No external developers or paid technical staff are required during the research phase. Consequently, the human effort involved in designing, developing, and testing the system is considered part of the academic learning process and does not incur financial cost.

Justification for Each Expense

Overall, the project is designed as a research prototype, which allows the use of freely available development tools and cloud service tiers. By utilizing open-source frameworks, existing computing resources, and student development effort, the total estimated cost for system development remains minimal. This cost-efficient approach allows research to focus on evaluating the system’s educational impact and technical feasibility without requiring significant financial investment.

1.Development Tools

Table 1-development of tools cost

Tool	Purpose	Cost
Flutter	Mobile app development	$0
Android Studio / VS Code	Coding environment	$0
Python (FastAPI / Flask)	Backend development	$0
Git / GitHub	Version control	$0

2.Firebase Services

Table 2-Firebase services cost

Feature	Free Limit
Authentication (Email / Google login)	Free
Firestore Database	1 GB storage
Database reads	50,000 per day
Database writes	20,000 per day
Storage	Small free quota

3.Final Budget Estimation

Table 3-final budget

Component	Cost
Flutter Development Framework	$0
Python Backend (FastAPI / Flask)	$0
Firebase Authentication	$0
Cloud Firestore Database (1GB free tier)	$0
Google Sign-In Integration	$0
Development Tools (VS Code / Android Studio)	$0
Application Publishing	$0 (not publishing)

Total Estimated Budget = $0

8. WORK BREAKDOWN STRUCTURE (WBS)

Task Breakdown per Member

The project is divided into four main components, with each team member responsible for developing and integrating a specific module of the system.

Sachintha Nirmal S.M – AI Chatbot and Learning Navigation Module
Responsible for designing and developing the AI-powered chatbot that assists students in understanding physics concepts. This includes implementing grade-adaptive responses, guiding students through lesson content, and integrating the chatbot with the application's structured learning navigation system. The member also designs the hierarchical structure of grades, lessons, topics, experiments, and activities to ensure students can easily access learning materials.

Wijerathna C D W – Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module
Responsible for implementing the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module. This includes monitoring student quizzes and virtual experiment performance, analyzing learning progress at sub-lesson and topic levels, detecting weak areas, generating adaptive personalized feedback, providing remediation resources for weak students, and implementing structured learning progression enforcement within the mobile virtual physics laboratory.

Vidurangi – Interactive Physics Experimentation and Learning Module
Responsible for developing interactive physics simulations and virtual experiments within the mobile application. This includes designing practical experiment activities where students can interact with virtual objects, measure physical outcomes, and understand physics calculations through guided explanations.

Arthanayake C T H – Intelligent Content Retrieval and Gamified Learning Generation Module
Responsible for implementing the intelligent content retrieval system and gamified learning features. This includes developing keyword-based search functionality for physics learning materials and creating 2D educational games aligned with lesson topics to improve student engagement and motivation.

Timeline Alignment

The project development process is organized into several phases aligned with the research timeline.

• **Phase 1 – Requirement Analysis and Literature Review**
Identification of research problems, review of existing studies, and definition of system requirements.

• **Phase 2 – System Design**
Design of system architecture, database structure, user interface layouts, and component interactions.

• **Phase 3 – System Development**
Implementation of each module by the respective team members, including mobile interface development, backend services, experiment simulations, and AI chatbot integration.

• **Phase 4 – Integration and Testing**
Integration of all system components, testing system functionality, verifying data flow between modules, and fixing identified issues.

• **Phase 5 – Evaluation and Documentation**
Evaluation of system performance, preparation of research documentation, and final report submission.

Balanced Workload Distribution

The project tasks are distributed among team members to ensure a balanced workload and effective collaboration. Each member is responsible for a major system component while contributing to system integration and testing activities. This modular development approach allows parallel progress in different parts of the system while maintaining coordination among team members. By assigning clearly defined responsibilities, the team ensures efficient development, effective knowledge sharing, and timely completion of the research project.

WBS Table

Table 4-WBS Table

WBS ID	Task Name	Member	Description	Duration
1.0	Project Initiation	All	Identify research problems, define project scope, and obtain supervisor approval	Month 1
1.1	Literature Review	All	Review research papers related to virtual labs, mobile learning, learning analytics, adaptive feedback, RAG-based chatbots, gamification, and personalized learning systems	Month 1 – Month 2
1.2	Research Proposal Preparation	All	Prepare research proposal including introduction, research gaps, objectives, and methodology for all four modules	Month 2
2.0	System Requirement Analysis	All	Identify functional and non-functional requirements of the complete mobile virtual physics laboratory system	Month 3
2.1	System Architecture Design	All	Design overall system architecture including Flutter mobile application, Python backend services, Firebase, and database structure	Month 3
2.2	Database Design	All	Design Cloud Firestore data structure for user profiles, quiz scores, experiment data, analytics records, chatbot data, game records, learning paths, and curriculum content	Month 3
3.0	User Authentication	Wijerathna C D W	Implement student registration, login, Google Sign-In, guest access, and student profile management using	Month 4

	and Profile Module		Firebase Authentication and Cloud Firestore	
3.1	Intelligent Quiz Evaluation System	Wijerathna C D W	Develop intelligently generated sub-lesson quizzes with MCQs and structured questions aligned to Grade 6–11 curriculum, with shuffled retake support	Month 5 – Month 6
3.2	Experiment Performance Evaluation	Wijerathna C D W	Implement system to evaluate practical experiment steps, record performance outcomes, and identify poorly performed experiment areas per sub-lesson	Month 6 – Month 7
3.3	Learning Analytics and Weakness Detection	Wijerathna C D W	Develop algorithms to analyze sub-lesson quiz and experiment results and detect weak theory and practical areas	Month 7 – Month 8
3.4	Adaptive Feedback Generation	Wijerathna C D W	Implement rule-based feedback mechanism to generate personalized feedback messages and display improvement suggestions on the student home page	Month 8
3.5	Personalized Learning Path Assignment	Wijerathna C D W	Develop learning path assignment system that detects weak students and assigns targeted resources including explanations, guided examples, experiment procedures, and video materials; enable reattempt after completion; assign additional external resources for persistent weakness	Month 8 – Month 9
3.6	Dashboard and Visualization Development	Wijerathna C D W	Develop student dashboard with sub-lesson performance indicators, quiz and experiment scores, weak area summaries, feedback messages, and learning path access	Month 9 – Month 10
4.0	Grade-Tagged Physics Knowledge Base	Sachintha Nirmal S.M	Build a grade-tagged physics knowledge base containing curriculum-aligned physics content organized by grade level, lesson, topic, and activity for RAG retrieval	Month 4 – Month 5
4.1	RAG-Based AI Chatbot Pipeline	Sachintha Nirmal S.M	Implement the Retrieval-Augmented Generation pipeline using LLM to retrieve relevant syllabus content and	Month 5 – Month 6

			generate physics explanations for student queries	
4.2	Grade-Adaptive Response Generation	Sachintha Nirmal S.M	Develop grade-adaptive response generation that tailors chatbot answers based on the student's grade level and lesson context	Month 6
4.3	Structured Learning Workflow Module	Sachintha Nirmal S.M	Design and implement the structured learning workflow including hierarchical navigation for grades, lessons, topics, experiments, and activities	Month 4 – Month 5
4.4	Chatbot Integration with Mobile Frontend	Sachintha Nirmal S.M	Integrate the AI chatbot interface and structured learning workflow screens into the Flutter mobile application	Month 7
5.0	Syllabus-Aware Keyword Search Engine	Arthanayake C T H	Implement keyword-based search engine that matches student-entered physics keywords with Sri Lankan National curriculum syllabus materials	Month 4 – Month 6
5.1	Adaptive 2D Game Generation Engine	Arthanayake C T H	Create curriculum-aligned 2D educational games using Unity that adapt game difficulty and content based on student grade level and lesson topic	Month 5 – Month 7
5.2	Grade-Adaptive Scoping Framework	Arthanayake C T H	Design scoping framework that maps digital learning activities and game difficulty to structured lesson-level objectives for Grades 6–11	Month 6 – Month 7
5.3	Content Retrieval and Game Integration	Arthanayake C T H	Integrate keyword-based retrieval system and game generation module with Firebase Firestore for NIE-aligned curriculum data storage and retrieval	Month 7 – Month 8
6.0	Virtual Experiment Simulation Development	Vidurangi H I D G	Develop interactive physics virtual experiments using Unity 2D where students can interact with virtual objects and perform guided experiments	Month 4 – Month 7
6.1	Measurement Extraction Module	Vidurangi H I D G	Implement real-time measurement extraction system using OpenCV and YOLOv8 to capture physics measurements such as distance, time, and velocity from student interactions	Month 5 – Month 6
6.2	Physics Calculation Module	Vidurangi H I D G	Apply physics formulas to extracted measurements to generate step-by-step	Month 6 – Month 7

			calculation results and concept explanations for students	
6.3	Experiment Performance Evaluation Module	Vidurangi H I D G	Implement scoring system that evaluates student experiment performance based on safety tool usage, apparatus selection, procedure steps, and final result accuracy	Month 7 – Month 8
7.0	System Integration	All	Integrate Flutter mobile application, Python backend analytics services, Firebase Authentication, Cloud Firestore, chatbot, experiment module, gamified learning, and analytics module into a complete unified system	Month 10 – Month 11
8.0	System Testing and Debugging	All	Conduct functional and performance testing of all modules including authentication, quiz evaluation, experiment evaluation, weakness detection, feedback generation, learning path assignment, chatbot accuracy, RAG retrieval, experiment simulations, and game functionality	Month 10 – Month 11
9.0	System Evaluation	All	Evaluate usability and effectiveness of the complete system including mastery level prediction accuracy, chatbot performance, RAG retrieval accuracy, experiment evaluation accuracy, search relevance, game engagement, and learning path effectiveness	Month 11
10.0	Documentation and Final Report	All	Prepare final research documentation including results, discussion, and complete system description for all four modules	Month 11 – Month 12
11.0	Final Presentation Preparation	All	Prepare presentation slides, demonstration materials, and final system submission	Month 12

9. GANTT CHART

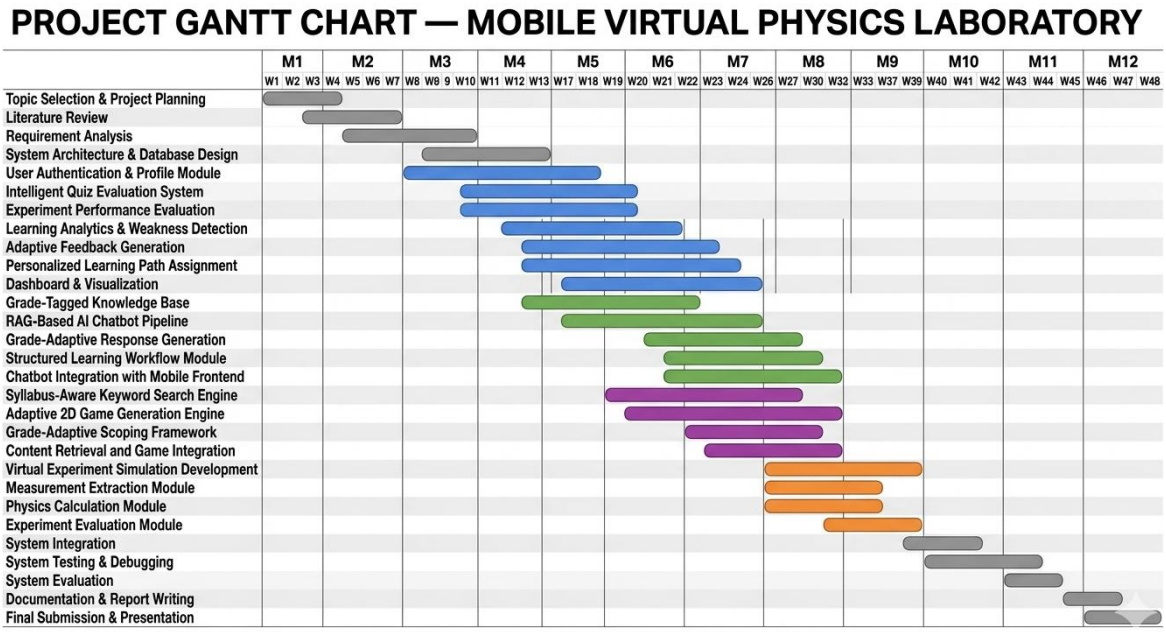


Figure 4-Gantt chart

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11.APPENDICES

[Appendix A — Survey instruments]

This research does not include survey-based data collection methods at this stage. Instead, the system evaluation is conducted through functional testing, user interaction testing, and performance analysis using the System Functionality Checklist. Therefore, no survey instruments are included in this study.

[Appendix B — Detailed Diagrams]

This section includes detailed diagrams related to the system design and methodology of the Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module. These diagrams provide a visual representation of the research workflow, module structure, and system data flow.

Key diagrams include:

- **Research Methodology Diagram** — illustrates the overall design science research approach used to develop the module
- **Research Methodology Process Diagram** — shows the step-by-step process flow from student login through quiz and experiment assessment, weakness detection, learning path assignment, reattempt, and escalating resource provision
- **High-Level System Architecture Diagram** — presents the three-layer architecture including the Flutter presentation layer, Python application logic layer, and Cloud Firestore data layer with all module components and their interactions
- **Module Architecture Diagram** — illustrates the five main parts of the module: user authentication, profile management, sub-lesson assessment, learning analytics and weakness detection, and personalized learning path assignment
-

These diagrams illustrate the flow of activities, system components, and interactions between the mobile application, Python backend, Firebase Authentication, Cloud Firestore database, and external resources such as YouTube videos and reference materials.

[Appendix C — Preliminary UI Mockups]

- Student login and registration screen
- Home page screen with feedback messages and suggestion section
- Sub-lesson selection screen showing lesson and sub-lesson structure
- Quiz attempt screen with MCQ and structured question layout
- Quiz result screen displaying quiz score and identified weak questions
- Virtual experiment screen with step-by-step guided interaction
- Experiment result screen displaying experiment score and weak step identification
- Student dashboard screen showing sub-lesson performance indicators and weak area summaries
- Personalized learning path screen showing assigned resources for weak sub-lessons including explanations, guided examples, experiment procedures, and video materials



Figure 5-UI sample mockups

[Appendix D — Risk Analysis]

Table 5-Risk analysis

Risk	Likelihood	Impact	Mitigation Strategy
Inaccurate weakness detection due to insufficient quiz and experiment data	Medium	High	Validate detection logic with sample student data during testing phase
Students bypassing learning path by not engaging with assigned resources	Medium	Medium	Enforce structured progression so students cannot advance without completing the current sub-lesson
Firebase free tier limits exceeded during testing with multiple users	Low	Medium	Limit concurrent test users during prototype evaluation stage
Video-based learning materials unavailable due to broken YouTube links	Medium	Medium	Use multiple backup links and verify all links before system evaluation
Python backend response delays affecting real-time feedback display	Low	High	Optimize analytics processing logic and test backend performance under load
Student quiz data loss due to network interruption	Low	High	Implement local caching in Flutter app and sync with Cloud Firestore on reconnection
Shuffled quiz retakes generating repeated questions	Low	Medium	Implement question bank with sufficient variation per sub-lesson to ensure unique sets

[Appendix E — Ethical Documents]

- Informed consent form for student participants involved in system usability testing
- Parental consent form for underage student participants (Grade 6–11)
- Data privacy and confidentiality agreement ensuring student learning data is used solely for research purposes
- Data handling policy outlining how student profiles, quiz scores, experiment results, and learning path records are stored, accessed, and protected in Cloud Firestore
- The system is designed to support fair and inclusive access to education without bias